PROJECT SUMMARY

GROUP 9

CASCADE VS DIRECT?

Understanding Accents
Beyond Borders



REGIONAL ENGLISH ACCENT ROBUSTNESS IN SPEECH TRANSLATION

DIRECT AND CASCADED ENGLISH-TO-CHINESE SYSTEMS

Motivation

Speech translation should work well for speakers with different accents. Earlier research reports accent differences in speech recognition.

Research Problem

How does translation quality differ across regional English accents in Direct and Cascaded systems?

Goal

Compare the systems using the same recordings and Chinese references.

Objectives

Compare quality

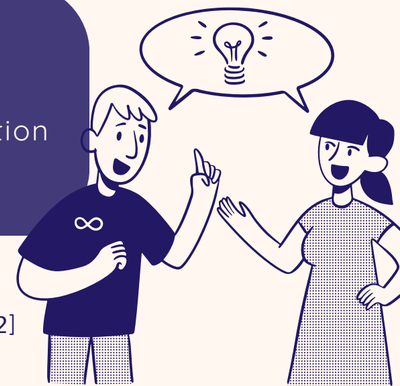
Direct vs Cascaded translation, overall and within each accent.

Measure accent differences

How scores and differences between systems vary across accents.

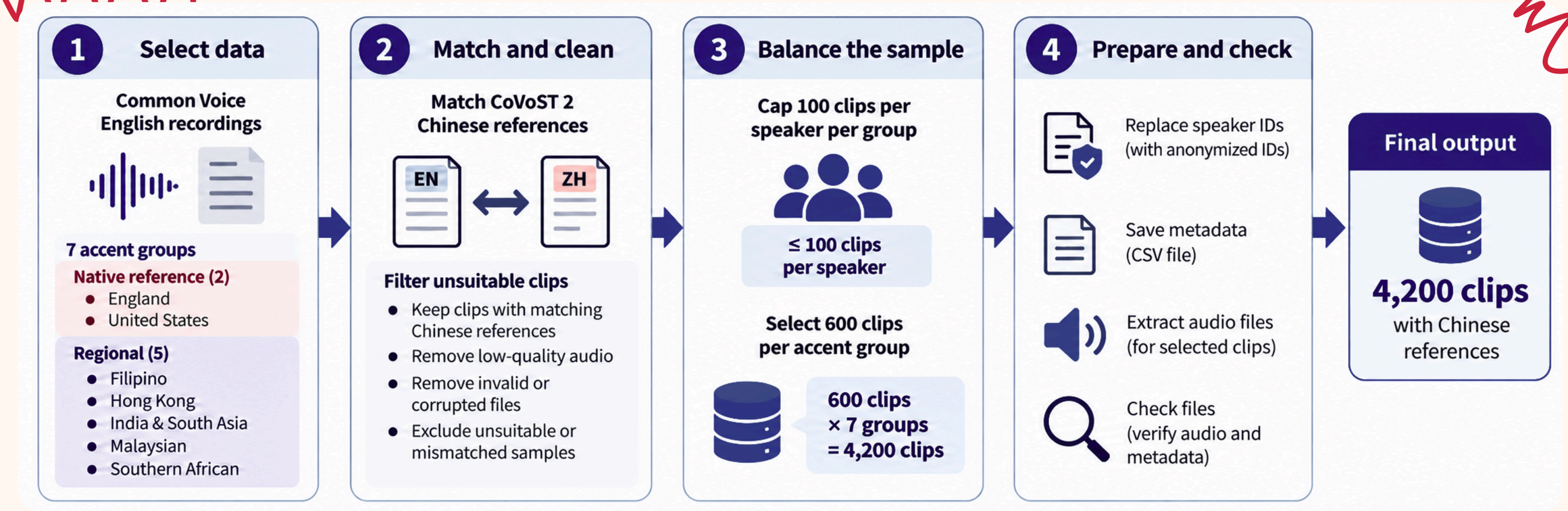
Examine transcription errors

How they relate to cascade translation quality



Accent ASR study [1]; Direct-Cascade ST comparison [2]

DATASET RECAP

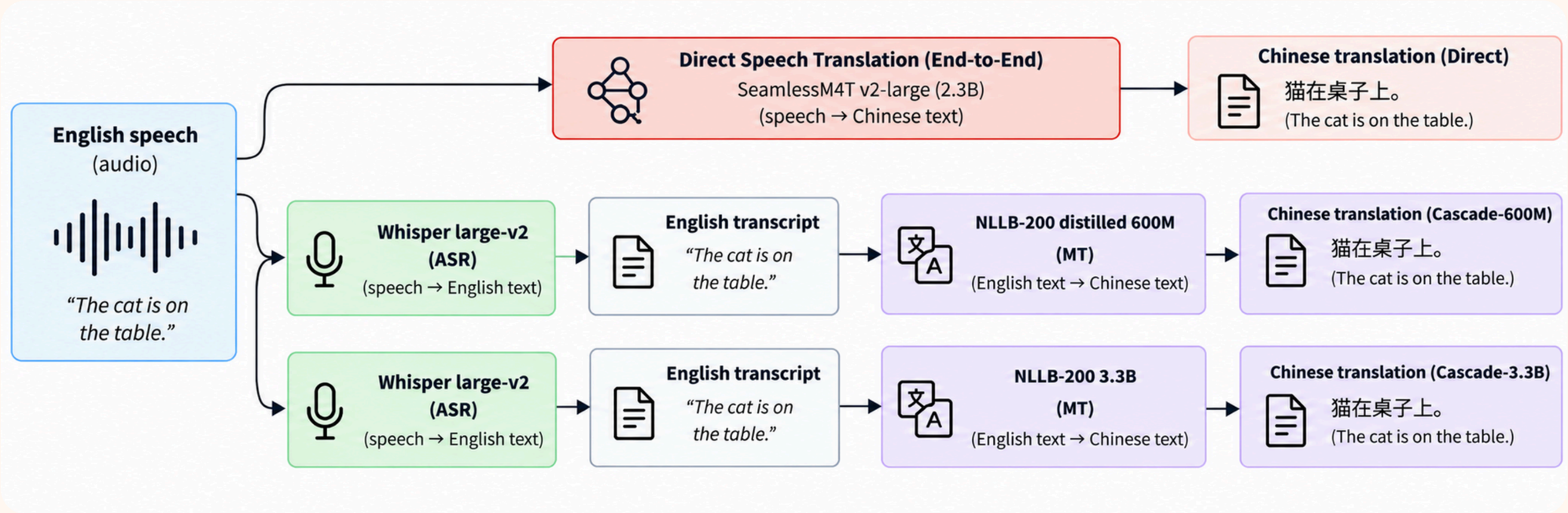


1586 Speakers | **5.39h** Of speech | **4.62s** Mean clip

Limit: the accent groups have different speakers and sentence content.

Common Voice [3]; CoVoST 2 [4]

THREE PRETRAINED SYSTEMS (STRONGER CASCADE)



- The larger translation model helps balance the comparison.
- Each cascade runs Whisper large-v2 separately, then uses its own NLLB model.

Note: the systems still differ in model size, training data, and design.

SeamlessM4T v2 [5]; Whisper [6]; NLLB [7]; related ST implementation [8]

EXPERIMENTAL DESIGN & IMPLEMENTATION RECAP

Design:
- evaluate pretrained models
- No training or tuning.

Scores:
- BLEU
- chrF and chrF++ for translation
- Word error rate (WER) for transcription.

Same settings:
- 16 kHz mono audio
- greedy decoding
- 256-token limit
- seed 760.

Hardware:
- Google Colab
- NVIDIA T4
- FP16

Runs	Inference time
Direct	65.0 min
Cascade-600M	65.1 min
Cascade-3.3B	**77.5 min**

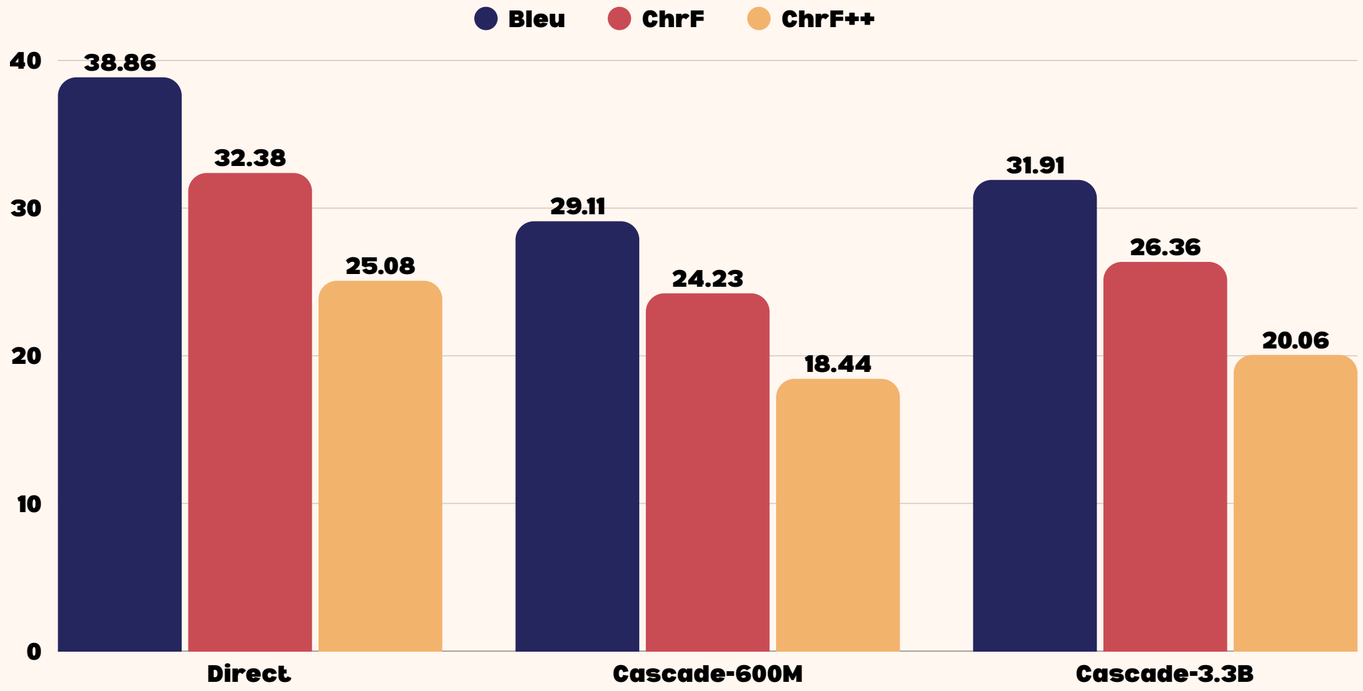
Our Project Github Repo Link

Implementation: Python and Google Colab notebooks for data preparation, Direct and Cascaded speech translation, evaluation, statistical analysis, and error investigation.

https://github.com/Nlhmmh/cs760-accent-st-robustness-ml-research

BLEU [9]; chrF [10]; chrF++ [11]; scoring settings [12]

OVERALL TRANSLATION QUALITY

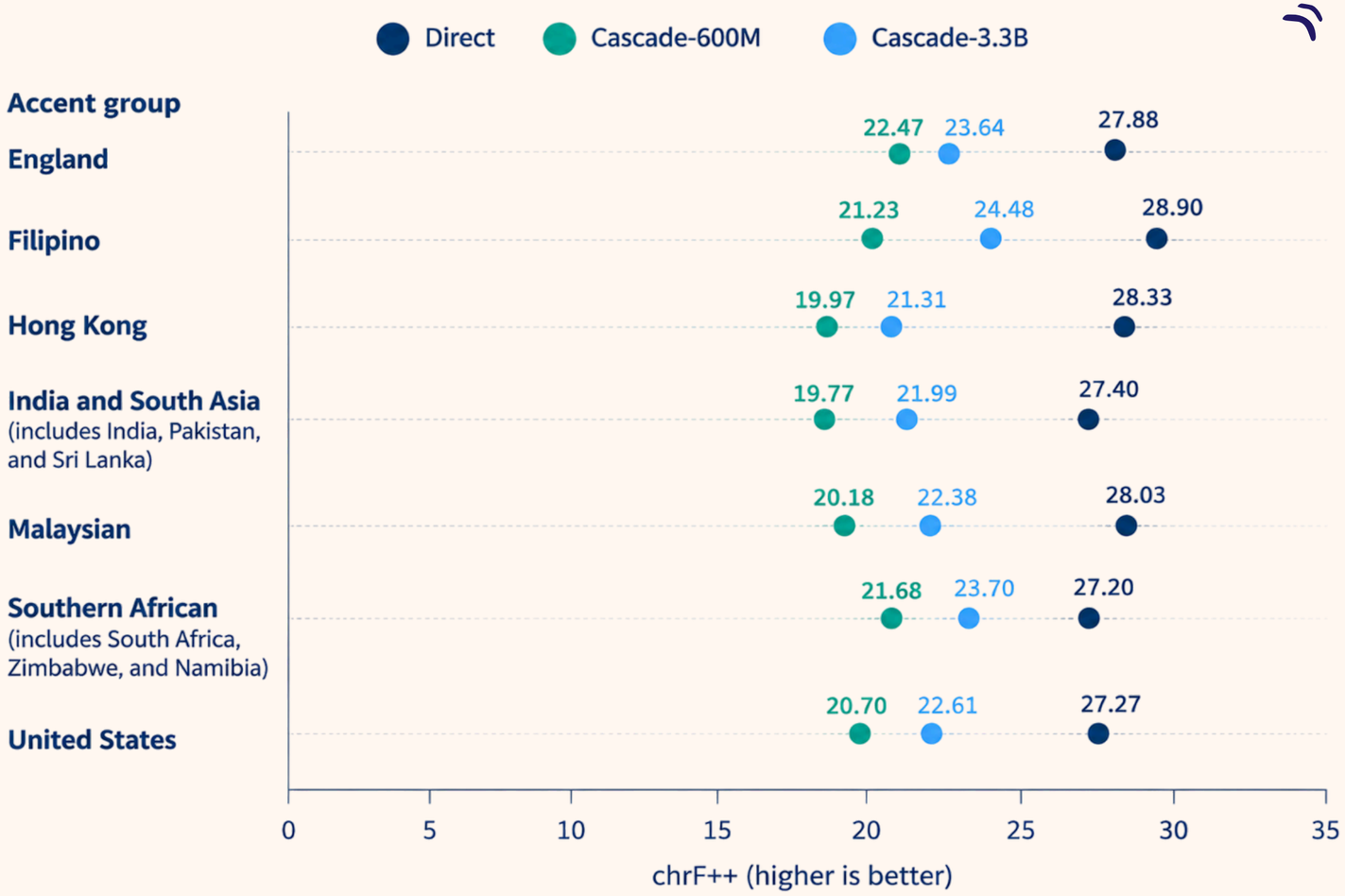


Key Findings:

- Cascade-3.3B improves BLEU by 2.81 points over Cascade-600M
- Direct leads Cascade-3.3B by 6.94 BLEU points
- All three metrics show the same ranking: Direct > Cascade-3.3B > Cascade-600M
- Scores are corpus-level results across 4,200 clips; higher is better

BLEU [9]; chrF [10]; chrF++ [11]; Notebook 8: Evaluation

TRANSLATION QUALITY ACROSS ACCENTS



Key Findings:

- Direct has the highest mean clip-level chrF++ in every accent group
- Cascade-3.3B outperforms Cascade-600M in all seven groups
- Paired score-gap intervals for Direct versus both cascades are above zero in all seven groups
- Accent rankings differ between systems

Limitation:
Dataset-specific results; self-reported accents; groups differ in speakers and content. Paired intervals are unadjusted for multiple comparisons.

chrF++ [11]; paired resampling [13]; Notebook 8: Evaluation

SCORE VARIATION ACROSS ACCENTS

Method

- Resample speakers within each accent 1000 times
- Clips and systems kept paired
- Spread = highest accent avg - lowest accent avg

Example: Direct's best accent is 28.90, its lowest 27.20, a spread of 1.71 (unrounded scores).

Finding

- Accent averages and rankings differ between systems. Statistical evidence for differences within each system is limited.
- Confidence intervals show uncertainty around each spread estimate. A maximum minus a minimum is never negative, so a positive lower bound is not evidence of an accent effect.

Accent Spread

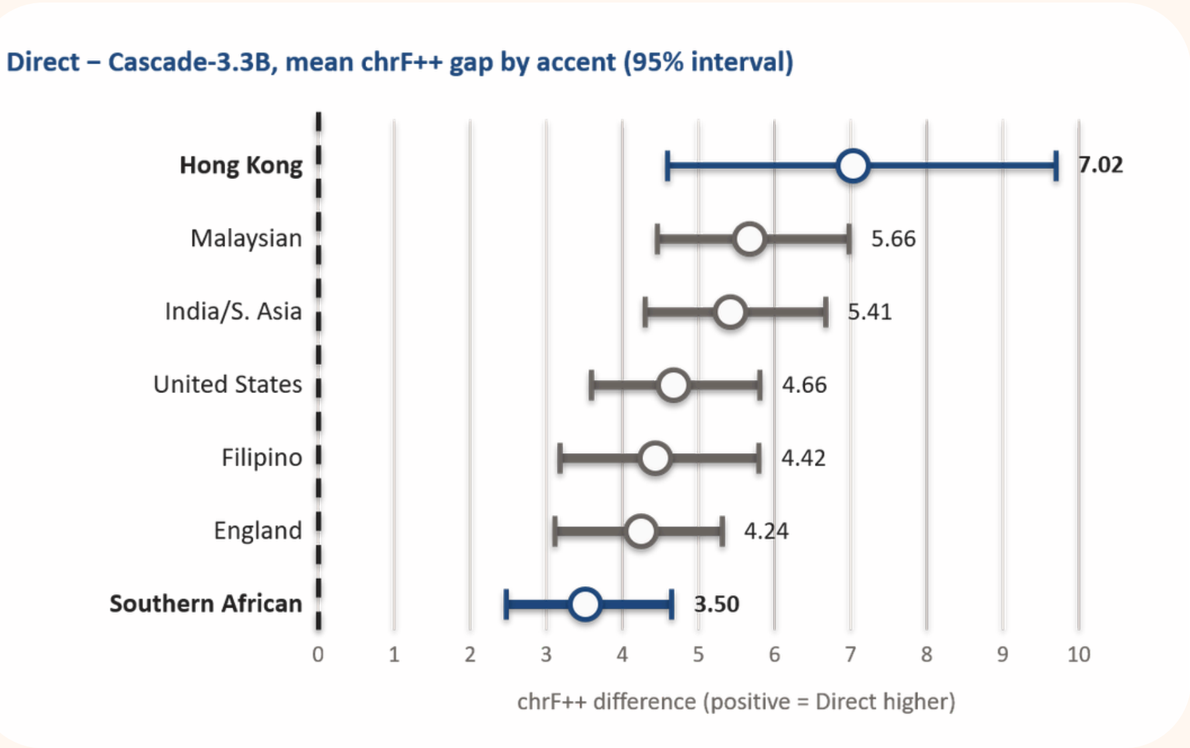
1.71	2.70	3.17
95% Interval : 1.43-5.06	95% Interval : 1.95-4.74	95% Interval : 2.03-5.67
Direct	Cascade 600M	Cascade 3.3B

Limitation

- **Groups have different speakers and sentences**
- **Observed, not proven accent effects**

chrF++ [11]; paired resampling [13].

ACCENT VARIATION ACROSS SYSTEM



Overall gap, 95% interval:

vs 600M: +7.00 (+6.45 to +7.60)
vs 3.3B: +4.99 (+4.39 to +5.70)

Finding

- Both Direct-gap intervals are above 0 in all 7 accent groups.
- Speaker-level tests support variation in the gaps, with small effects.

Limitation

- **Per-accent intervals are not corrected for multiple comparisons. Model family, size, and training data also differ.**

chrF++ [11]; paired resampling [13].

WHAT DRIVES THE GAP?

Q5 · The <unk> marker

+5.42

Direct's lead over 3.3B without <unk> clips (+4.99 with them)

- In 12.1% of Direct outputs, similar in every accent

Q6 · ASR errors

3.6 → 8.2

Direct's lead over 3.3B: perfect transcript → at least one wrong word

- 70% of clips have no ASR error
- WER vs 3.3B score: Spearman −0.20, all 7 accents

Q7 · Bigger MT model

+2.01

3.3B over 600M in every accent (95% interval 1.61 to 2.42)

- **No accent dependence (p = 0.17)**
- **Lifts the level, not the accent pattern**

Associations on this dataset. The systems differ in family, size and training data.

Mean sentence chrF++; notebooks 09 and 10 [13].

WHY, AND WHAT WOULD HELP?

Q8 · Likely causes, strongest first

1. ASR errors carried into the cascade: the main accent driver
2. Idioms that no literal translation can match
3. <unk> on rare words: about 1.2 points, same in every accent
4. Bigger MT model: about +2 in every accent

Example: "There you have it!" → reference 给你！(here you go); the literal 这就是它! scores 0

3.61

Direct's lead over 3.3B even with a perfect transcript. Still unexplained: model design, training data or reference style?

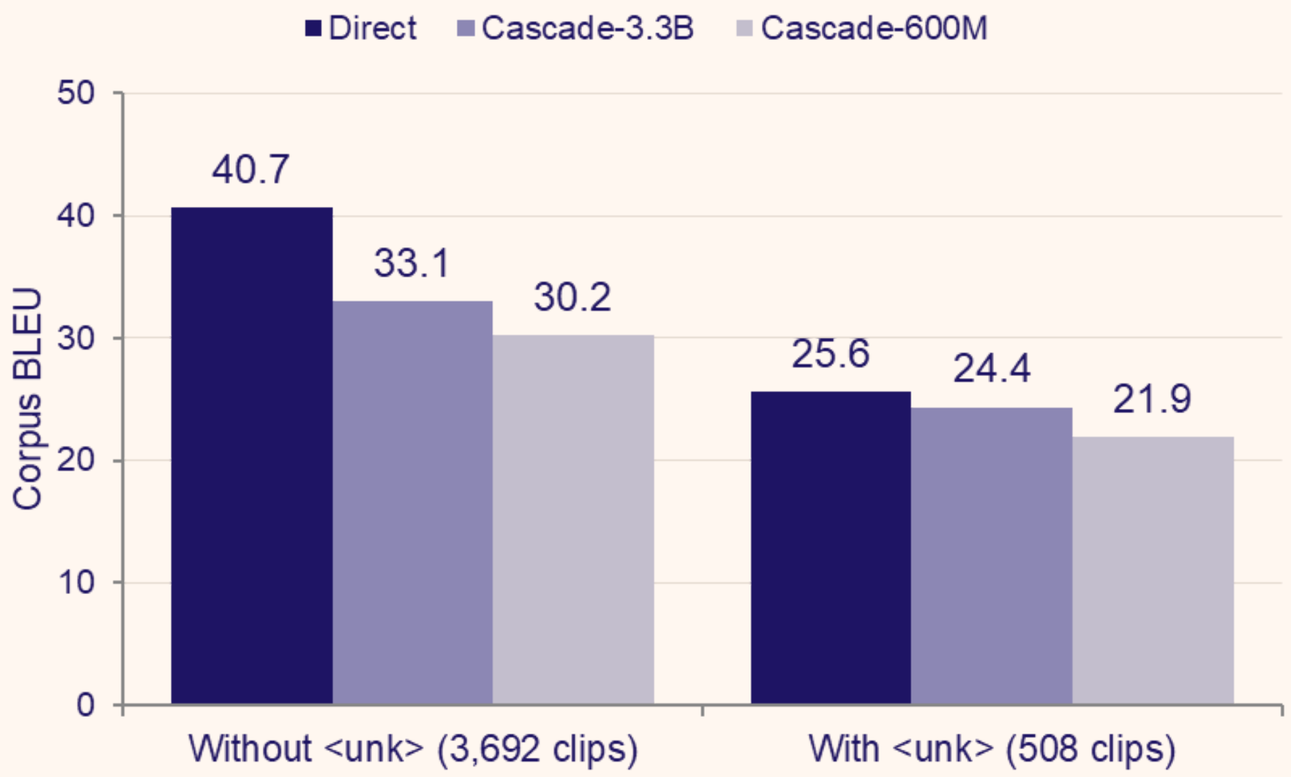
Q9 · What would help?

Try first	**Accent-robust ASR for the cascade: rerun in about 78 GPU minutes on a T4**
Explored	**Blocking <unk> while decoding: BLEU fell from 28.2 to 15.5**
Explored	**Cascade output on <unk> clips: chrF++ fell from 25.08 to 24.89**
Next	**COMET for idioms; one clip per speaker already holds (4.69)**

Notebooks 06, 07 and 10; unk_analysis results.

IS DIRECT REALLY BETTER, OR PARTLY BROKEN?

Question from last Presentation : Direct BLEU is 25.64 on the 508 clips with <unk> and 40.67 on the other 3,692. What about the cascades?



Direct is ahead

On this evaluation set, over the 3.3B cascade: +6.9 BLEU on all clips and +7.6 on the clips without <unk>.

But the lead shrinks

On the 508 clips with <unk>, the lead over 3.3B is +1.3 (a point estimate).

All systems score lower

The cascades drop 8.3 to 8.7 BLEU on these clips; Direct drops 15.0.

Clips are grouped by whether Direct's output contains <unk> (the cascades never do). The two groups contain different sentences, so the gap is not the cost of the marker.

BLEU [9]; scoring settings [12]

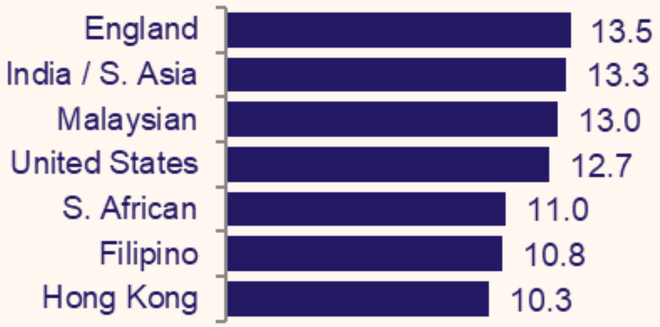
WHAT WE FOUND ABOUT THE <UNK>

English sentence	Direct output (speech)	Runner-up token	P("<")	P(runner-up)
Where did you find that apple?	你在哪里找到那个<unk>果?	苹果	0.56	0.27
Please bring soap.	请带上肥<unk>.	皂	0.66	0.24
Garlic can be used to fight vampires.	大<unk>可以用来对抗吸血鬼.	蒜	0.63	0.16

In these examples the marker interrupts an everyday Chinese word. In text-input decoding, an appropriate Chinese token was the runner-up to "<". Probabilities come from text input, not speech input.

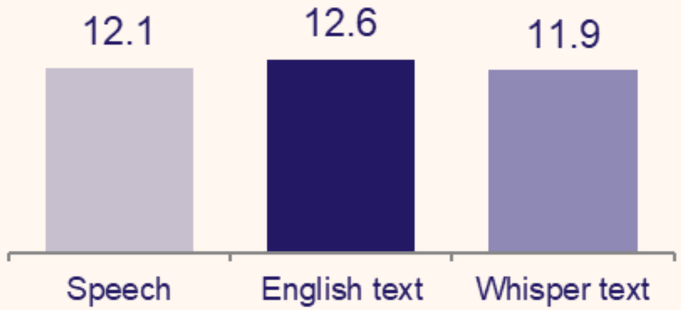
1. No clear accent association

10.3 to 13.5% by accent; speaker-clustered p = 0.50; small differences remain possible



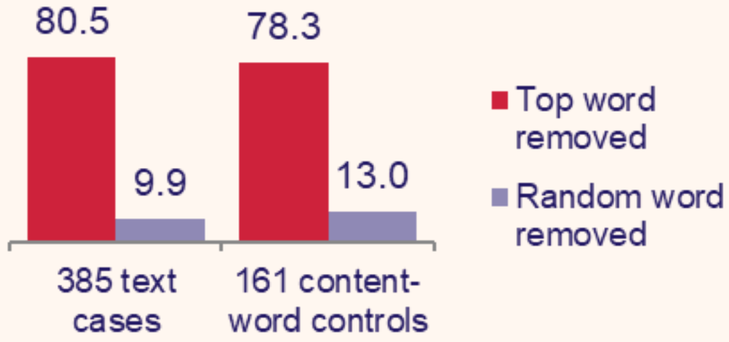
2. Audio not required

Text input also produces the marker: 528/4,200 clips (12.6%), including 449 of the 508 speech-affected clips.



3. Linked to specific words

Text input, 385 cases: removing the top word removed the marker in 80.5% vs 9.9% for a random word



Not fixed: on 35 affected clips, blocking the spelling gave "<unk >" variants (BLEU 28.2 → 15.5); beam search removed the exact marker in only 1 of 35.

Post-hoc diagnostics; original predictions unchanged. Check 3: % of sentences where the marker disappears (removing a word changes the meaning).

WHAT WORKED, WHAT WE LEARNED, WHAT TO IMPROVE

What worked & What we learned

What worked

All 3 systems were scored on the same 4200 clips and references, so differences are paired. Saved predictions let us recompute scores and intervals without rerunning the models.

Account for repeated speakers

579 speakers give several clips (up to 88). For the overall Direct–600M mean sentence chrF++ gap, the 95% clip-bootstrap CI is about 19% narrower than the speaker-bootstrap CI: [6.54, 7.47] vs [6.45, 7.60].

Test metrics on Chinese

SacreBLEU's default tokenizer gives 0.43; the zh tokenizer gives 38.86. Corpus and mean-sentence chrF++ differ (25.08 vs 27.86).

Define every ID and number

sample_id is a speaker ID, not a clip key. One result has two valid numbers (80.5% and 78.3%) because the control sets differ.

Limitation → next step

Speakers and sentences differ across accents

→Account for both dependencies together. Use the 309 shared sentences (693 clips) for a limited within-sentence comparison.

Single reference; models differ

→ One reference per clip: add a learned semantic metric or bilingual human evaluation.
→ Different model sizes and training data: limit conclusions to these systems.

Unresolved <unk> mechanism

→ Blocking its spelling lowered BLEU on 35 clips. Test blocking the start tokens "<" and "__<"; check translation quality and new errors.

Sources: repository README, intervals_long.csv, unk_analysis results, runs/evaluation_results/merged_3system_with_metrics.csv.

chrF++ [11]; scoring settings [12]; paired resampling [13]; NLLB [7]

CONCLUSIONS AND CONTRIBUTIONS



Quality:
Direct scored highest in all 7 accent groups. Overall corpus BLEU: 38.9, vs 31.9 (3.3B) and 29.1 (600M).

Accents:
best-to-worst gaps are 1.7 to 3.2 mean sentence chrF++ points; Direct's lead: 3.5–7.0 over 3.3B, 5.4–8.4 over 600M.

Transcription errors:
a weak negative association with cascade scores (rho −0.19, −0.20); Direct's mean sentence chrF++ lead over 3.3B is 3.6 when normalised WER is 0 and 8.2 when it is positive: an association, not a causal estimate.

<unk>:
12.1% of Direct outputs, no clear accent association; the mechanism is unresolved.

Member	Contribution
Arizona Xing	Evaluation dataset (4,200 clips); further <unk> analysis; Survey section on speech translation architectures.
Nathan Naylin	Pipelines and runs; earlier <unk> tests; documentation; Organised and integrated the literature survey.
Patricia Jennesha	Three-system evaluation and score table; Survey section on accent robustness in speech translation.
Zhitong Li	Speaker-level statistics; shared metric table; Survey section on accent robustness in speech recognition.
Jidni Mayukh	Metrics and statistics review; Q1–Q9 synthesis; Survey sections on evaluation metrics and statistical reliability.

Code repository: github.com/Nlhmmh/cs760-accent-st-robustness-ml-research

BLEU [9]; scoring settings [12]

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[2] L. Bentivogli et al., “Cascade versus direct speech translation: Do the differences still make a difference?,” in Proc. 59th Annu. Meeting Assoc. Comput. Linguistics and 11th Int. Joint Conf. Natural Language Processing (ACL-IJCNLP), vol. 1, 2021, pp. 2873–2887, doi: 10.18653/v1/2021.acl-long.224.

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[9] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in Proc. 40th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2002, pp. 311–318, doi: 10.3115/1073083.1073135.

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GenAI Use

ChatGPT/Codex assisted with auditing, structure and wording.

the team is responsible for the design, implementation, verification and conclusions.

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THANK YOU!

THOUGHTS
& AND QUESTIONS?

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